INCH-POUND

MIL-I-82819(OS)
5 February 1990
SUPERSEDING
WS 20889A
9 June 1986
(See 6.7)

MILITARY SPECIFICATION

INSULATION, THERMAL, POLYISOPRENE, SILICA AND ASBESTOS FILLED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for one type of calendered, uncured thermal insulation composed of polyisoprene elastomer with silica and asbestos fillers, referred to herein as thermal insulation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378 Plastic Film (Polyethylene Thin Gage)

RR-S-366 Sieves, Standard for Testing Purposes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 214	Sieve, Analysis of Granular Metal Powders
ASTM D 129	Sulfur in Petroleum Products by the Bomb Method
ASTM D 412	Rubber Properties in Tension
ASTM D 792	Specific Gravity (Relative Density) and Density of Plastics by Displacement
ASTM D 1646	Testing Rubber from Natural or Synthetic Sources--Viscosity and Vulcanization Characteristics (Mooney Viscometer)
ASTM D 2240	Rubber Property--Durometer Hardness
ASTM D 3280	Analysis of White Zinc Pigments
ASTM E 28	Softening Point by Ring-and-Ball Apparatus
ASTM E 203	Water Using Karl Fischer Reagent
ASTM E 324	Relative Initial and Final Melting Points and the Melting Range of Organic Chemicals

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

QUEBEC ASBESTOS MINING ASSOCIATION

Chrysotile Asbestos Test Manual

(Application for copies should be addressed to the Quebec Asbestos Mining Association, 580 Grande Allee East, Suite 320, Quebec, P.Q., Canada, G1R 2K2.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the thermal insulation shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2 **Material**. The material shall be silica and asbestos filled polyisoprene rubber consisting of the ingredients, the formulations, and the subsequent manufacture into uncured thermal insulation as described herein.

3.3 **Ingredients**.

3.3.1 **Formulation**. The properties of each lot of ingredients used in the thermal insulation formulation shall conform to table I (see 6.2).

TABLE I. Thermal insulation.

Ingredient	Parts by Weight	
	Minimum	Maximum
Polyisoprene elastomer (Natsyn 2200)	98	102
Asbestos, Chrysotile (Grade 4T-30)	98	102
Silica hydrate (Hisil 233)	4.9	5.1
Zinc Oxide (99 percent pure, minimum)	4.9	5.1
Sulfur (97 percent pure, minimum)	2.45	2.55
Styrenated phenols (VANOX 102) (previously known as Agerite Spar)	0.98	1.02
Polymerized trimethyl dihydroquinoline (Agerite Resin D)	0.49	0.51
Hydrofol Acid	1.96	2.04
N-tert butyl 1-2-2 benzothiazolesulfenamide (Santocure NS)	1.47	1.53

3.3.2 **Properties of ingredients**. The properties of each lot of ingredients used in the thermal insulation formulation shall conform to table II (see 6.2).

3.4 **Thermal insulation**.

3.4.1 **Material and process specification**. The contractor shall prepare a specification, or an equivalent document, defining preparation of the insulating material. The document shall include, but not be limited to, composition, procedures, and methods for handling and weigh out of ingredients, mix sequence, calendering, transfer, and storage of non-vulcanized insulation, time, temperature and humidity limits, the equipment used, and processes and quality controls. Capability of the process and equipment shall be determined by the conformity of the thermal insulation to the requirements of table III.

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Table II. Constituent materials.

Ingredient	Property	Unit	Value		Test Methods
			Minimum	Maximum	
Polyisoprene Elastomer	—	—	—	—	Quebec Asbestos Mining Association
Asbestos, Chrysotile	Grade Classification	Textural Number	4T30	—	4.5.14.1
Silica Hydrate	Surface area	Meters^2/gram	137	167	ASTM E 203
	Water content	Percent	—	10	
Zinc Oxide	Sieve analysis	Percent through a No. 325-mesh sieve	99	—	4.5.14.2
	Purity	Percent	99	—	ASTM D 3280 using a diphenylamine indicator
Sulfur	Sieve analysis	Percent through a No. 325-mesh sieve	97	—	4.5.14.3
	Purity	Percent	98	—	ASTM D 129
Styrenated Phenole	Density	Mg/m^3	1.06	1.10	T9D Vanderbilt
Polymerized Trimethyl Dihydroquinoline	Softening point	°C	74	—	ASTM E 28
Stearic Acid	Melting point	°F	130	—	ASTM E 324
N-tert Butyl 1 2-2 Benzothiazolesulfenamide	Melting point	°C	104	—	ASTM E 324

TABLE II. Constituent materials.

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TABLE III. Chemical, physical, and thermal properties (cured insulation).

Property	Value	
	Minimum	Maximum
Cured Material		
Specific gravity	1.35	1.39
Tensile strength, with grain, psi	1350	---
Tensile strength ratio, with grain/across grain	1.5	2.1
Ultimate elongation, with grain, %	40	---
Hardness, Shore A	78	88
Uncured Material		
Ash content, %	43.0	46.0
Volatile content, %	---	1.5
Mooney Viscosity, ML 1+4 at 212°F	Report	

3.4.1.1 **Process approval.** The document defining the composition and detailed processes selected by the contractor shall be identified with an assigned number and submitted to the contracting activity for approval. Any change in the compounding process after first article inspection (see 3.1) shall be submitted to the contracting activity for written approval. The supplier shall incorporate approval changes and reidentify the document reflecting the changes. Requests for changes shall be accompanied by supporting data.

3.4.2 **Chemical, physical, and thermal properties.** The chemical, physical, and thermal properties of the calendered thermal insulation shall conform to table III.

3.5 **Safety.**

3.5.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.6 **Calendered sheets.** The calendered stock shall conform to 3.6.1 through 3.6.4.

3.6.1 **Thickness.** The thickness of the calendered stock shall be as specified in the contract or purchase order (see 6.2). The tolerance of the calendered stock thickness shall conform to the requirements shown in table IV.

TABLE IV. Thickness tolerance of calendered sheet.

Nominal thickness, inch	Allowable tolerance, inch
Less than 0.020	± 0.003
0.021 to 0.029	± 0.004
0.030 to 0.049	± 0.005
0.050 to 0.069	± 0.006
0.070 or greater	± 10 percent

3.6.1.1 **Roll width.** Unless otherwise specified in the purchase document, the width of the roll shall be 44.0 ± 1.0 inches.

3.6.2 **Laminate.** The calendered stock exceeding 0.030 inch in thickness shall be laminated in accordance with table V.

TABLE V. Laminate.

Nominal thickness, inch	Minimum number of laminae
0.031 to 0.100	2
0.101 to 0.300	3

3.6.3 **Polyethylene film carrier.** The uncured sheet stock shall be calendered onto a smooth polyethelene film, and a second smooth polyethylene film shall be applied to the opposite side of the final calendered pass. The polyethylene film may be locally perforated, if required, to release air entrapment which has occurred during the calendering operation. The polyethylene film shall be a minimum of 0.004 inch thick, shall be of a contrasting color to the sheet stock, and shall conform to L-P-378, Type I, Class 1, Grade C, Finish 1.

3.6.4 **Roll weight.** The weight of a roll of uncured calendered stock shall not exceed 275 pounds.

3.7 **Age prior to shipment.** The age of the material from date of final calendering to shipping shall not exceed 30 days.

3.7.1 **Temperature control.** The calendered stock shall be shipped in a temperature environment not higher than 90°F. Between date of calendering, and date of shipping, excursions above 90°F are permissible provided the temperature

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does not exceed 120°F and does not exceed a cumulative total of 24 hours above 90°F (see 5.1.1).

3.7.2 **Storage temperature**. The calendered stock shall be stored in a temperature environment of 40°F maximum, except for a maximum cumulative excursion of 15 days above 40°F and up to 100°F. Any excursion above 100°F is unacceptable.

3.8 **Shelf life**. The shelf life of the calendered stock shall be 6 months from date of manufacture when stored as specified in 3.7.2.

3.8.1 **Shelf life extension**. The shelf life expiration date of an individual lot may be extended one time for an additional 3-month period provided the material, upon retest, conforms to table VI, except that the retest for ash content, volatile content, and specific gravity are not required for storage life extension.

3.9 **Defect criteria**. The cumulative area of the rubber in each roll with defects such as wrinkles, voids, delaminations, etc., shall not exceed 10 percent of the net area of the rubber in the roll.

3.10 **Workmanship**. The material shall be protected from exposure to direct sunlight and free from surface contamination such as grease, oil, foreign materials, and other defects which could render the thermal insulation unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection**. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance**. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations**. The contractor shall have the toxicological product formulations and associated information available for

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review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspection (see 4.4). Failure to meet the requirements specified herein shall be cause for rejection of the first article sample.

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table VI.

TABLE VI. Quality conformance inspection.

Examination and Tests	Requirement	Test Method
Visual examination	3.3.1, 3.3.2, 3.4.1, 3.6.3, 3.10 and section 5	4.5.2
Unvulcanized		
Ash content	3.4.2	4.5.9
Volatile content	3.4.2	4.5.10
Mooney viscosity	3.4.2	4.5.13
Vulcanized		
Specific gravity	3.4.2	4.5.4
Tensile strength	3.4.2	4.5.5
Ultimate elongation	3.4.2	4.5.6
Ratio of tensile strengths	3.4.2	4.5.7
Hardness	3.4.2	4.5.8

4.4.1 **Sampling.**

4.4.1.1 **For roll weight.** Sampling for roll weight shall be each roll in the lot.

4.4.1.2 **For polyethylene film application and packaging.** Sampling for visual examination to determine conformance to 3.6.3 and section 5 shall be each roll in the lot.

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4.4.1.3 **For visual examination.** Sampling for visual examination shall consist of one random sample of sufficient size to perform the quality conformance tests.

4.4.2 **Lot.** A lot shall not exceed 1250 pounds of uncured calendered thermal insulation.

4.4.2.1 **Lot, crossblended.** A lot may consist of more than one batch provided it is thoroughly crossblended.

4.4.2.2 **Lot, no crossblend/batch.** A batch of unvulcanized thermal insulation is that quantity of material obtained from one charge of mill final batch and shall be traceable to an internal mixer (Banbury type). Without batch crossblending a batch is a lot. A lot may be one batch or crossblend of multiple batches.

4.4.2.3 **Traceability.** Traceability records shall be included as part of the certification package (see 6.2) and shall provide a complete lot/batch or identification history, by serial number, from ingredient lot numbers (see 3.2) through contracting activity stock and lot numbers. The following shall be required:

a. Ingredient lot numbers.

b. Master batch serial number(s).

c. Final batch serial number(s).

d. Crossblended lot numbers.

e. Calendered material (contractor).

(1) lot number (same as crossblended lot number).

(2) roll numbers, weight, and stock thickness.

f. Calendered material (contracting activity).

(1) stock number and lot number.

(2) identify all other material from single manufacturer's lot by contracting activity stock and lot numbers.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

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NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercially nonreagent grade materials.

4.5.1 **Sample preparation.** Test specimens for specific gravity, tensile strength, elongation, and hardness determination shall be prepared from calendered stock cured at 250 ± 5°F for 120 ± 10 minutes under a pressure of 1000 psi minimum using 0.105-inch molds to provide 0.100- to 0.110-inch-thick material for test specimens. Prior to cutting test specimens to the required width, the test specimen material temperature shall be from 60 to 90°F for a minimum of 24 hours. Test specimen material shall be cut to test specimen width using a 0.50-inch-wide die. Test values shall be recorded as the average of a minimum of five test specimens. Retesting may be done on other comparable (five specimens minimum) groups, however, the number of retests shall not exceed two, and all values shall be recorded.

4.5.2 **Visual examination.** Samples selected in accordance with 4.4.1.3 shall be visually examined to determine compliance with 3.6.3, 3.10, and section 5. The temperature recorder (see 5.1.2) shall be examined to determine conformance to 3.7.1. Contractor certification verifying conformance to 3.3.1, 3.3.2, and 4.4.2.3 shall be required (see 6.2).

4.5.3 **Thickness.** Thickness shall be determined using a calibrated micrometer or visually by counting the number of laminae.

4.5.4 **Specific gravity.** Specific gravity shall be determined in accordance with ASTM D 792. The average specific gravity at 25.0 ± 0.5°C shall be recorded to the nearest 0.001.

4.5.5 **Tensile strength.** Tensile strength shall be determined in accordance with ASTM D 412, except that specimens shall be prepared with the grain in the long direction and the average shall be tested using an Instron tester with an integral recording chart. The tester shall be set at 200 pounds full scale and shall use a 20-inch-per-minute crosshead and chart speed.

4.5.6 **Ultimate elongation.** Ultimate elongation shall be determined in accordance with ASTM D 412, using the same specimens as in 4.5.5, except that the average shall be recorded to the nearest 1 percent.

4.5.7 **Ratio of tensile strength.** The ratio of tensile strengths shall be performed by preparing and running specimens perpendicular to the grain direction and using the method described in ASTM D 412. The two averages (tensile strength parallel to the grain and tensile strength perpendicular to the grain) shall be recorded to the nearest 0.01.

4.5.8 **Hardness.** Hardness shall be determined in accordance with ASTM D 2240, except that the average shall be recorded to the nearest whole scale reading.

4.5.9 **Ash content.** The ash content shall be determined as follows:

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a. Weigh duplicate specimens, 6 to 8 grams, to the nearest 0.001 gram into a tared crucible.

b. Ignite under a fume hood using a Meeker burner.

c. Place the specimens in a muffle furnace at 1500 ± 50°F for a minimum of 60 minutes.

d. Remove, reweigh, and calculate the ash content as follows:

$$\text{Percent ash} = \frac{W2 - W}{W1 - W} \times 100$$

Where: W1 = weight of specimen and crucible prior to heating, g

W2 = weight of specimen and crucible after heating, g

W = weight of crucible, g

e. Record the average of the duplicate specimens to the nearest 0.1 percent.

4.5.10 **Volatile content**. The volatile content shall be determined as follows:

a. Weigh duplicate specimens, 10 to 15 grams, to the nearest 0.001 gram.

b. Place the specimens in a vacuum oven at 107 ± 3°C (225 ± 5°F) with an absolute pressure of 0 to 3 inches of mercury for 4 hours minimum.

c. Remove, reweigh, and calculate the volatile content as follows:

$$\text{Percent volatile content} = \frac{(B - A) \times 100)}{B}$$

Where: A = weight of specimen after heating, g

B = weight of specimen prior to heating, g

d. Record the average of the duplicate specimens to the nearest 0.1 percent.

4.5.11 **Roll weight**. Roll weight shall be determined using scales of applicable accuracy.

4.5.12 **Age control**. Age control shall be verified by the date of calendering marked on the packaging and the date of shipment on the shipping paper of the material.

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4.5.13 **Viscosity characteristics.** Mooney viscosity of the elastomer shall be determined by shearing disk viscometer method in accordance with ASTM D 1646 and the following:

a. Operate the viscometer in accordance with ASTM D 1646 using the large rotor and a 60 second preheat.

b. Testing temperature shall be 212°F.

c. The Mooney viscosity shall be the torque in inch-pounds 4 minutes after starting the motor.

d. Record test results.

4.5.14 **Properties of ingredients.**

4.5.14.1 **Surface area of silica hydrate.** Surface area of silica hydrate shall be measured by surface absorption of monomolecular layers of gas using either a Numinco-Orr Surface Area-Pore Volume Analyzer or a Perkins-Elmer Shell Sorptometer or similar instruments. Follow the instructions for calibration and measurement as recommended by the instrument manufacturer.

4.5.14.2 **Sieve analysis of zinc oxide.** Sieve analysis of zinc oxide shall be determined in accordance with ASTM B 214. Only a No. 325-mesh sieve shall be used.

4.5.14.3 **Sieve analysis of sulfur.** Sieve analysis of sulfur shall be determined, using a sieve conforming to RR-S-366, in accordance with the following:

a. Place a 50- to 100-gram sample, weighed to the nearest 0.1 gram, on a No. 325-mesh sieve in a Tyler Ro-Tap machine and shake for a minimum of 20 minutes.

b. Weigh the residue on the sieve to the nearest 0.05 gram.

c. Calculate as follows:

Sieve analysis, percent passing through a 325-mesh sieve

$$= 1 - \frac{A}{B} \times 100$$

Where: A = weight of residue on sieve, g

B = weight of sample, g

4.6 **Inspection of packaging.** The containers and container markings shall be visually examined to verify compliance with section 5.

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5. PACKAGING

5.1 **Preservation and packaging.** Preservation and packaging of the thermal insulation shall be such as to prevent exposure to direct sunlight, ultraviolet radiation, ozone, and water. Container interior finishes or coatings shall be of such a nature as to prevent contamination or physical damage of the thermal insulation. The thermal insulation shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.1.1 **Shipping time and environment.** The maximum allowable shipment time from date of shipment to date of delivery shall be specified by the contract or purchase order. The insulation material shall be shipped in a temperature environment not greater than 90°F. Excursions above 90°F are permissible provided the temperature does not exceed 120°F and the time does not exceed a cumulative total of 24 hours above 90°F.

5.1.2 **Temperature recorder.** A temperature recorder shall be placed with the material to record the temperature during all handling, storage, and transportation. The recorder shall be capable of producing a continuous record with an accuracy of 6°F.

5.2 **Packing.** The thermal insulation, packaged as specified in 5.1, shall be packed in wood shipping containers with wood center supports in a manner to ensure safe delivery to destination.

5.3 **Marking.**

5.3.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture, mixing date, crossblend date, calendering date.

d. Shelf life and inspection/test date for extension (see 3.8.1). In addition, the marking shall include the following:

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"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 3-MONTH PERIOD PROVIDED THE THERMAL INSULATION, UPON RETEST, CONFORMS TO THE VISUAL EXAMINATION, MOONEY VISCOSITY, TENSILE STRENGTH, ULTIMATE ELONGATION, RATIO OF TENSILE STRENGTHS, AND HARDNESS REQUIREMENTS OF MIL-I-82819."

e. Storage condition labels of 40°F maximum.

f. SHIPPING TEMPERATURE SHALL NOT EXCEED 90°F.

g. Total lot weight.

h. Number of rolls in lot.

i. Toxicity warning (if applicable).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The thermal insulation is intended for use in the manufacture of Standard Missile Mk 104 Rocket Motors.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Material Safety Data Sheets Required (see 3.5.2 and 6.4).

d. Whether first article inspection is required (see 3.1 and 6.3).

e. DOD FAR 52.246-15 (see 3.3.1, 3.3.2, 4.4.2.3, and 4.5.2).

f. Thickness of calendered stock (see 3.5.1).

g. Special marking, if other than as specified in 5.2.2.

h. Traceability records (see 4.4.2.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a

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standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Possible material.** The following materials have been found to meet the requirements of this specification. This is only given for information and is not restrictive.

Ingredient	Trade Name	Supplier
Polyisoprene elastomer	Natsyn 2200	Goodyear Tire and Rubber Company
Asbestos, Chrysotile	Grade 4T-30	Johns Manville Products Corporation
Silica hydrate	Hisil 233	Johns Manville Products Corporation
Zinc Oxide	Laccotine	Los Angeles Chemical
Styranated phenols	T-9-D	R.T. Vanderbilt and Company
Polymerized trimethyl dihydroquinoline	Agerite Resin D	R.T. Vanderbilt and Company
Hydrofol Acid	Stearic Acid	Sherex Co.
N-tert Butyl 1-2-2 benzo-thiazolesulfenamide	Santocure NS	Monsanto Industrial Chemicals Co.

6.6 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

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6.7 **Supersedure information.** MIL-I-82819 incorporates the following engineering change proposals (ECPs), notice of revision (NOR) and specification change notices (SCNs):

ECP		NOR		SCN	
SSS1035	9/19/84			SCN 1	9/19/84
MTA037	11/30/87			SCN 6	12/11/87
				SCN 7	12/11/87
ARC011	4/25/89	ARC011.1	4/25/89		

Preparing Activity:
Navy - OS
(Project 1340-N803)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-82819	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
INSULATION, THERMAL POLYISOPRENE, SILICA AND ASBESTOS FILLED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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